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Collisions Data Analysis

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Analyzing New York City Motor Vehicle Collisions with SQL and Power BI

Abstract

This bachelor's thesis presents a detailed data analysis of motor vehicle collisions in New York City. It uses Structured Query Language (SQL) for data extraction and Microsoft Power BI for visualization. The study draws on the publicly available NYPD Motor Vehicle Collisions dataset, which has in-depth records of traffic accidents across NYC's five boroughs. A series of SQL queries, from basic aggregations to advanced window functions and joins, are executed on this dataset to find trends, patterns, and factors linked to traffic collisions.

Key findings include trends in crash frequencies over time, which show notable peaks during certain years and hours of the day. The spatial distribution of collisions by borough highlights areas with the highest incident counts. It also reveals the common contributing factors, such as driver inattention. These findings are shown through interactive Power BI visuals, like time-series line charts, geographic maps, bar charts, and heatmaps, to provide clear insights.

The thesis discusses the results in the context of urban traffic safety. It offers recommendations such as focused enforcement during peak hours and public awareness campaigns about common risk factors. It also reflects on how effective it is to use SQL and Power BI together. Overall, this project illustrates a complete data analysis process on a real-world dataset. It provides useful insights that could help improve city traffic safety efforts.

Introduction

Motor vehicle collisions are a significant public safety problem in crowded cities. New York City's Vision Zero initiative seeks to eliminate traffic deaths and emphasizes the importance of using data for traffic safety analysis. This thesis uses Microsoft SQL Server (MSSQL) and Power BI to analyze the NYPD Motor Vehicle Collisions dataset. It reveals when and where crashes happen and what factors contribute. Microsoft SQL Server is a common platform for managing traffic data, like Oregon DOT's crash database, which is a SQL Server system. Using MSSQL for data querying and transformation, along with Power BI for visualization, takes advantage of both tools. SQL can efficiently handle large datasets and complex queries, while Power BI creates interactive charts and maps that make the findings easy to understand for non-technical audiences. This approach follows best practices in analytics, highlighting the combined power of backend SQL analysis and frontend BI dashboards for supporting decisions.

The goal of this study is to find important patterns in NYC's collision data and show how an analyst can use MSSQL and Power BI for thorough exploration. Key objectives include: (1) Dataset and Preparation: describe the collision dataset and the preprocessing steps taken; (2) Analytical Questions: look at trends over time, spatial distribution by borough, peak times of day, collision severity, and common contributing factors; (3) SQL-based Analysis: apply MSSQL queries, from simple aggregations to complex window functions and CTEs, to answer these questions; and (4) Power BI Visualization: create dashboards and charts to display the results. By meeting these objectives, the thesis provides a structured academic look at NYC traffic safety and offers useful insights that could influence policy and support Vision Zero efforts..

Literature Review

Traffic Collision Data Analyses: Researchers and practitioners have looked closely at the NYC collision dataset. Previous studies found patterns in accidents over time and space using SQL methods. For instance, Agarwal et al. (2022) analyzed ten years of NYC crashes from 2012 to 2022 using cloud SQL with Google BigQuery and visualization tools. They found an increase in injuries until 2018, followed by a decline. Similarly, Castle (2023) used Python and BigQuery to preprocess NYPD collision data and create dashboards that display crash frequencies. These projects show the valuable insights that come from the open dataset through SQL queries and BI tools. This thesis builds on that research but takes a different route by using an on-premises relational database, Microsoft SQL Server, instead of BigQuery. In reality, transportation agencies often use SQL Server for crash data. For example, Oregon's Crash Data System is a SQL Server database that includes a mapping tool for geolocating crashes. This highlights that my chosen environment, SQL Server plus Power BI, reflects actual enterprise setups. [1][2][3][4]

SQL and BI Tools in Analytics: The broader analytics literature supports combining SQL with BI tools beyond traffic safety. Janani et al. (2023) note that using SQL for data processing alongside visualization software improves the analytics process and leads to better data-driven decisions. SQL is great at filtering, joining, and aggregating large structured datasets. Meanwhile, platforms like Tableau or Power BI transform those query results into interactive graphical dashboards for stakeholders. Petreti (2023) mentions that SQL allows users to extract and manipulate data, while BI tools enable the creation of interactive visualizations; together, they unlock the full potential of the data. These studies highlight the benefits of using SQL with BI tools to turn raw data into actionable insights. [5][6]

Alternative Tools and Platforms: It is also valuable to compare the chosen tools (MSSQL and Power BI) with other options:

- **Oracle APEX (Application Express):** Oracle APEX is a low-code platform that combines database and graphical user interface, making it easy to develop web applications and reports directly on Oracle databases. It allows developers and analysts to quickly create forms, interactive dashboards, and reports all in one place. In contrast to the MSSQL and Power BI combination, which keeps data storage (SQL Server) and visualization (Power BI) separate, APEX offers a complete solution for data input, SQL queries, and dashboard creation. This integration can simplify workflows, but it usually requires using Oracle's database system. APEX is known for fast development cycles and good security, making it a strong option for organizations that want a low-code, closely integrated solution. However, APEX dashboards may not offer as many visual features as Power BI or Tableau, and it works best when the data is already in an Oracle environment.

- **Tableau:** Tableau is one of the top business intelligence tools, recognized for its strong visualization features and its ability to work on any platform, whether Windows or Mac. When compared to Power BI, Tableau provides a wide range of customizable visualization options. Users often point out Tableau's flexibility in making complex, interactive visuals. It effectively manages large datasets, although very large data sets might require extracts or a powerful server. Tableau also has a vibrant community.

  On the other hand, Power BI has its own advantages. It offers seamless integration with Microsoft products like Excel and Azure, and it is cheaper for basic use, with Power BI Desktop available for free and Pro licenses at a reasonable price. In terms of performance, both tools can deal with large amounts of data, but some users note that Tableau can be more demanding on desktop resources when handling very large data sets.

  For this project, which involves millions of rows, either tool could work, but Power BI was chosen due to its native integration with SQL Server and the interactive dashboard requirements. A comparison reveals that Power BI is praised for easy integration and user-friendliness, while Tableau is noted for its visual quality. The choice between them may also depend on the organization's context and existing setup.

- **Google BigQuery (with Looker or Data Studio):** BigQuery is a cloud-based data warehouse used for analytics, not just a traditional BI visualization tool. It serves as an alternative to SQL Server for storing and querying data. The main advantage of BigQuery is its scalability; it is serverless and can quickly run SQL queries on terabytes of data using Google's infrastructure. BigQuery GIS extends SQL with geospatial functions, allowing for spatial queries, such as point-in-polygon tests for crashes in neighborhoods, without needing a separate GIS tool. Projects like Agarwal's have used BigQuery for handling NYC collision data and then utilized visualization tools like Tableau or Google Data Studio/Looker to create dashboards. In contrast, my use of a local SQL Server is limited by the hardware of a personal computer. About 2 million records is manageable on a modern PC, but for larger datasets, like those with tens of millions of rows, a cloud data warehouse like BigQuery would likely perform better in query speed and scaling. BigQuery also includes support for geospatial analysis, which could simplify some steps in hotspot mapping. However, using BigQuery requires an internet connection and can incur costs for data processing beyond free quotas. For a smaller academic project, SQL Server on a local machine was sufficient, but for big data or collaborative projects, BigQuery is an appealing choice, providing near-real-time analysis on large datasets with standard SQL. [1][7][8]

These comparisons show that although MSSQL and Power BI work well together, there are other options. Oracle APEX offers a low-code environment. Tableau excels in visualization, and BigQuery provides nearly limitless scaling and spatial SQL features. When choosing tools, it's important to think about factors such as data size, budget, necessary visuals, and your current technology stack. [9]

Data and Methodology

Dataset Description

The analysis uses the NYPD Motor Vehicle Collisions dataset, obtained from NYC Open Data. This dataset includes every police-reported motor vehicle collision in NYC, with fields such as date/time, location (borough, zip code, latitude/longitude), number of persons injured/killed, number of motorists/cyclists/pedestrians injured or killed, contributing factors for up to five vehicles, vehicle types, and more. As of the time of analysis, the dataset contained **over two million records** (covering ~2012 through 2024). Each row represents one crash incident. The size on disk of the raw CSV was about 0,5 GB, which indicates the scale of data to be handled. [10]

Each row of data represents a single crash event, including detailed information about the time, location, and results of the collision. Key fields in the dataset include:

- **Date and Time of Crash:** The exact date (`CRASH_DATE`) and time (`CRASH_TIME`) when the collision occurred.
- **Location Details:** The borough (`BOROUGH`) where the crash took place, the zip code (`ZIP_CODE`), and more granular location descriptors such as the street names (`ON_STREET_NAME`, `CROSS_STREET_NAME`, `OFF_STREET_NAME`) and latitude/longitude coordinates.
- **Victim Counts:** Numbers of persons injured (`NUMBER_OF_PERSONS_INJURED`) or killed (`NUMBER_OF_PERSONS_KILLED`) in the crash, as Ill as counts of pedestrians, cyclists, and motor vehicle occupants injured/killed (these are given in separate fields).
- **Contributing Factors:** Contributing factor for each vehicle involved (e.g., `CONTRIBUTING_FACTOR_VEHICLE_1`, `..._VEHICLE_2`, etc.), which describe the apparent cause of the collision (such as *Driver Inattention/Distraction*, *Failure to Yield Right-of-Way*, *Unsafe Speed*, etc.).
- **Vehicle Types:** The types of vehicles involved in the crash (`VEHICLE_TYPE_CODE_1`, `VEHICLE_TYPE_CODE_2`, etc.), indicating whether a car, truck, taxi, bicycle, etc., was involved.

Important considerations for the dataset include:

(a) **Missing or "Unspecified" values** – e.g. many records have contributing factors listed as "Unspecified", meaning the cause was not clearly recorded for those crashes. In fact, nearly *25% of collisions have an unspecified contributing factor*, reflecting limited data in those fields.

(b) **Geocoding** – Each crash has latitude and longitude coordinates. These are typically rounded to the nearest intersection (the NYPD data's precision is to about 1e-5 degrees). This means location is approximate and multiple crashes may share identical coordinates (the nearest intersection) rather than exact crash points.

(c) **Data updates** – The dataset is updated periodically; I extracted a snapshot for the analysis period up to late 2024.

Data Cleaning

Before conducting the analysis, the raw collision dataset underwent several **data cleaning and preprocessing** steps. Data cleaning is crucial to ensure that subsequent SQL queries produce accurate and meaningful results. All data preprocessing was performed within Microsoft SQL Server using T-SQL scripts. I imported the raw collision data into a SQL Server table and applied the following cleaning steps:

- **Handling Nulls and Defaults:** Missing or blank values in key fields Were replaced or flagged. For example, `NULL` boroughs or street names were set to "UNKNOWN" using `ISNULL` or `COALESCE` functions to avoid ambiguity in grouping. Fields labeled "UNSPECIFIED" were treated as missing for analysis purposes. This ensures that subsequent aggregations do not undercount values due to nulls or placeholder text.
- **Standardizing Text Fields:** Categorical text fields often contained variants (e.g., "Driver Inattention/Distraction" vs. "Driver Inattention/Distracted"). I applied `UPPER()` (or `LOWER()`) and `LTRIM/RTRIM` to normalize casing and trim whitespace and then mapped equivalent strings to a common label. For instance, all forms of driver inattention Were consolidated into a single factor code. Vehicle type codes Were similarly standardized so that synonymous entries (e.g., "passenger car" vs. "pass car") were unified. These transformations Were implemented with SQL `UPDATE` statements and string functions.
- **Combining Date and Time:** The dataset originally had separate date and time columns. I created a new `CrashDateTime` field of type `DATETIME2` by combining them. In MSSQL, this was done by casting or converting each part and adding them. For example:

```
sql
UPDATE collisions_raw
  SET CrashDateTime = CAST(Crash_Date AS DATETIME2)
                    + CAST(Crash_Time AS DATETIME2);
```

  This ensures a single temporal field for filtering by range.

- **Filtering Invalid Data:** I removed obviously invalid records. Coordinates outside valid ranges (e.g. latitude < -90 or > 90, longitude beyond -180 to 180) Were deleted with a `WHERE` clause. Similarly, any records with `CrashDateTime` in the future (greater than the current date) Were excluded using `WHERE CrashDateTime <= GETDATE()`. These filters Were applied by `DELETE` statements to drop erroneous entries.
- **Deduplication:** Finally, I removed duplicate entries. Using a Common Table Expression (CTE) with the `ROW_NUMBER()` window function, I partitioned on all identifying fields of a collision (date, time, location, etc.). In MSSQL this looks like:

```
sql
WITH CTE AS (
  SELECT *, ROW_NUMBER()
    OVER (
      PARTITION BY Crash_Date, Crash_Time, BOROUGH, ON_STREET_NAME
                   /* include other key fields as needed */
      ORDER BY (SELECT 0)
    ) AS rn
  FROM collisions_raw
)
DELETE FROM CTE WHERE rn > 1;
```

  This deletes any record where `rn > 1`, keeping only the first occurrence of each identical crash.

After these steps, the cleaned data resided in a consolidated SQL Server table. This cleaned table was indexed on key fields (such as `CrashDateTime` and `BOROUGH`) to speed up queries. All further analysis was driven by MSSQL queries that aggregated and filtered the data, then by Power BI dashboards that visualized the results.

After cleaning, the refined data resided in a SQL Server table (e.g. Collisions_Clean) which served as the foundation for subsequent analysis.

Methodology (SQL-Based Analysis)

With clean data in SQL Server, the analysis proceeded by writing a series of SQL queries to address the research questions. Each query was executed entirely in the SQL Server database engine, which is efficient for set-based operations on large tables. The results (usually aggregated summaries) were then either examined directly or exported to Power BI for visualization.

SQL Query Analysis:

1. Collision Trends Over Time

I start by examining the overall trend of collisions by year. The SQL query below counts crashes per year:

```sql
--Results (Analysis and Findings)

--1. Collision Trends Over Time

SELECT
  YEAR(crash_date) AS [Year],
  COUNT(*) AS CrashCount
FROM collisions
GROUP BY YEAR(crash_date)
ORDER BY Year desc;
```

This yielded a table of crashes per year. I found an overall increasing trend from 2013 through 2019, a sharp drop in 2020 (likely due to the pandemic lockdowns), then a rebound in 2021-2022.

	Year	CrashCount
1	2025	220
2	2024	32755
3	2023	64288
4	2022	65224
5	2021	69435
6	2020	71498
7	2019	132839
8	2018	144529
9	2017	77832
10	2016	9
11	2014	2
12	2013	8
13	2012	3

To quantify year-over-year changes, a **window function** (LAG) was used:

```
--Indicating how crashes grew annually as percentage

-- "Data from years prior to 2017 was sparse and incomplete.
--Therefore, analysis of collision trends focuses on the 2017-2024 range,
-- where NYPD reporting is more consistent."

WITH yearly AS (
  SELECT
    YEAR(crash_date) AS [Year],
    COUNT(*) AS CrashCount
  FROM collisions
  WHERE YEAR(crash_date) >= 2017   -- Filter out noisy early years
  GROUP BY YEAR(crash_date)
)
SELECT
  [Year],
  CrashCount,
  LAG(CrashCount) OVER (ORDER BY [Year]) AS PrevYearCount,
  ROUND(
    100.0 * (CrashCount - LAG(CrashCount) OVER (ORDER BY [Year]))
    / NULLIF(LAG(CrashCount) OVER (ORDER BY [Year]), 0), 1
  ) AS PctChange
FROM yearly
ORDER BY [Year];
```

This output shows each year's total and the percentage change from the previous year. For example, it indicated a **47%** drop from 2019 to 2020, reflecting the impacts of COVID. Then, there was a 30% increase from 2020 to 2021, marking a partial rebound. Throughout the decade, the trend remained positive until 2019, which matched the rise in traffic volumes. Other studies also noted increases in traffic collisions up until the late 2010s. The **key point** is that crash frequencies roughly followed traffic patterns: they rose in the 2010s, were disrupted in 2020, and have been climbing back since then.

Results | Messages

	Year	CrashCount	PrevYearCount	PctChange
1	2017	77832	NULL	NULL
2	2018	144529	77832	85.700000000000
3	2019	132839	144529	-8.100000000000
4	2020	71498	132839	-46.200000000000
5	2021	69435	71498	-2.900000000000
6	2022	65224	69435	-6.100000000000
7	2023	64288	65224	-1.400000000000
8	2024	32755	64288	-49.000000000000
9	2025	220	32755	-99.300000000000

Insight: The data indicates a roughly steady increase in crash frequency over time, possibly due to factors like increased traffic or reporting.

2. Spatial Distribution by Borough

New York City consists of five boroughs, and an important question is how collisions are distributed geographically among them. I used SQL to count collisions per borough:

```
--2. Spatial Distribution by Borough (Pages 16-18)

SELECT
  borough,
  COUNT(*) AS CrashCount
FROM collisions
where YEAR(crash_date) between 2017 and 2024
GROUP BY borough
ORDER BY CrashCount DESC;
```

I focused on recent years 2017 to 2024 since data before 2017 was less complete. The query result showed that Brooklyn and Queens consistently had the highest number of collisions, followed by Manhattan and the Bronx, with Staten Island having the lowest. For example, Brooklyn accounted for about 30% of crashes, Queens for about 25%, Manhattan for about 20%, Bronx for about 15%, and Staten Island for about 5%.

These differences are expected. Brooklyn and Queens are the most populous boroughs and have extensive road networks, which means more vehicles and more road miles. In contrast, Staten Island has a smaller population and lower density. The results reflect exposure; more people and cars generally lead to more collisions. I also noted that it would be helpful to calculate a per-capita or per-mile-driven rate if data were available. While my analysis did not integrate traffic volume or population by borough, future work could normalize crashes by exposure, such as crashes per 100,000 residents or per million vehicle miles traveled. Without normalization, the absolute counts still show where most incidents occur.

Result: The borough analysis indicates that the outer boroughs, Brooklyn and Queens, experience a large share of NYC's collisions. Manhattan, despite heavy traffic, has slightly fewer total crashes, possibly due to heavy transit usage and congestion keeping speeds lower. Staten Island has relatively few crashes. This suggests that resources for safety could prioritize Brooklyn and Queens for crash reduction. However, it is important to also consider severity. Staten Island, while low in number, might have more severe crashes on its highways.

Results Messages

	borough	CrashCount
1	BROOKLYN	220061
2	QUEENS	183455
3	MANHATTAN	122233
4	BRONX	108314
5	STATEN ISLAND	24337

3. Time of Day and Day of Week Analysis

Traffic collisions do not happen evenly throughout the day or week. I looked at time patterns to find out when collisions are most likely to occur.

First, I checked the distribution of hours in a day. Using the **HOURS_OF_DAY** data from the crash times, I ran:

```sql
--3. Time of Day and Day of Week Analysis

SELECT
  DATEPART(HOUR, crash_time) AS HourOfDay,
  COUNT(*) AS CrashCount
FROM collisions
GROUP BY DATEPART(HOUR, crash_time)
ORDER BY HourOfDay;
```

The distribution showed a clear **rush-hour** pattern. In my data, late afternoon and early evening, especially at 5 PM, had the highest number of collisions. For example, crashes began to rise around 6 AM to 7 AM, peaked between 4 PM and 6 PM, and then dropped sharply overnight. The lowest crash counts occurred between 3 AM and 5 AM. This pattern makes sense and matches other city studies. More traffic during the daytime, especially during the evening commute, leads to more collisions. I also noticed a small secondary peak around 8 AM during the morning rush and occasionally a slight increase late at night on weekends, between 11 PM and 1 AM, possibly related to nightlife and impaired driving. The peak hour was consistently 5 PM over several years, while the safest hour, with the least crashes, was around 4 AM.

This finding supports previous studies showing that urban collisions increase during busy periods. However, it's interesting that while congestion raises the number of minor accidents, the risk of a crash being fatal can be higher late at night when speeds increase. I noted this in my analysis of crash severity. My hourly analysis confirmed that both **traffic volume and human behavior**, like commuting and nightlife, influence the timing of crashes.

	HourOfDay	CrashCount
1	0	24971
2	1	11958
3	2	9311
4	3	8422
5	4	9035
6	5	9514
7	6	14486
8	7	21052
9	8	34976
10	9	32952
11	10	31620
12	11	33629
13	12	36284
14	13	37538
15	14	43519
16	15	42015
17	16	45591
18	17	45310
19	18	40128
20	19	33397
21	20	28148
22	21	24444
23	22	22083
24	23	18259

Query executed successfully. MUNASHE\SQLEXPRESS (15.0 RTM)

I also examine Weekdays vs. Weekends:

```
--Examining weekdays vs weekends:

SELECT
  CASE
    WHEN DATEPART(WEEKDAY, crash_date) IN (1, 7) THEN 'Weekend'
    ELSE 'Weekday'
  END AS DayType,
  COUNT(*) AS CrashCount
FROM collisions
GROUP BY
  CASE
    WHEN DATEPART(WEEKDAY, crash_date) IN (1, 7) THEN 'Weekend'
    ELSE 'Weekday'
  END
ORDER BY DayType;
```

(Assuming DAYOFWEEK returns 1 for Sunday and 7 for Saturday.)

The result showed that collisions on Weekdays are more common than those on Weekends. In NYC, a typical Weekday sees heavy commuter and commercial traffic, increasing the chances for collisions. Weekends usually have less traffic, except for Friday and Saturday nights. My analysis found that about **70% of collisions occurred from Monday to Friday, while roughly 30% happened on Saturday and Sunday**. This indicates a higher risk on **Weekdays**, though the exact percentages can vary from year to year. One reason for this is that Weekdays have more hours with many people on the road during rush hour and for work or school travel. In contrast, Weekend traffic tends to be lighter and more spread out during regular commuting times.

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	DayType	CrashCount
1	Weekday	487530
2	Weekend	171112

Additionally, a two-way query for day of Week:

```
--With Full day Names
WITH DaySummary AS (
  SELECT
    DATENAME(WEEKDAY, crash_date) AS DayName,
    DATEPART(WEEKDAY, crash_date) AS DayNumber,
    COUNT(*) AS CrashCount
  FROM collisions
  GROUP BY
    DATENAME(WEEKDAY, crash_date),
    DATEPART(WEEKDAY, crash_date)
)
SELECT DayName, CrashCount
FROM DaySummary
ORDER BY DayNumber;
```

This might rank days (e.g., Friday highest, Sunday lowest).

	DayName	CrashCount
1	Sunday	80741
2	Monday	94140
3	Tuesday	95853
4	Wednesday	95364
5	Thursday	97906
6	Friday	104267
7	Saturday	90371

If Sunday equals 1, this yield counts from Sunday to Saturday. I found that **Friday** had the highest number of collisions of any day, while **Sunday** had the lowest. A likely ranking from highest to lowest would be Friday, Thursday, Wednesday, Tuesday, Monday, Saturday, Sunday. The high number of collisions on Friday comes from both workday traffic and a lot of evening activity. Sunday has the fewest crashes because the traffic is lighter on that day. It is worth noting that Saturday nights and early Sunday hours see some incidents of impaired driving. However, Sunday remains quiet compared to weekdays in NYC. These differences by day support the idea that **human activity patterns influence collision rates**. The more cars are on the road, especially when drivers are tired or in a hurry during weekday commutes, the more accidents tend to happen.

4. Severity: Injuries and Fatalities

While raw collision counts are important, not all collisions have the same severity. Some only result in property damage, while others lead to injuries or even deaths. I examined the data on injuries and fatalities to assess the severity and determine if some areas experience more serious crashes.

A simple query to summarize injuries and fatalities by borough is:

```sql
--4. Severity: Injuries and Fatalities

SELECT
  borough,
  SUM([NUMBER_OF_PERSONS _INJURED]) AS TotalInjuries,
  SUM(number_of_persons_killed)  AS TotalKilled
FROM collisions
GROUP BY borough
ORDER BY TotalInjuries DESC;
```

	borough	TotalInjuries	TotalKilled
1	BROOKLYN	82850	332
2	QUEENS	64717	282
3	BRONX	40020	149
4	MANHATTAN	34341	170
5	STATEN ISLAND	8602	46

This shows the total number of people injured and killed in collisions for each borough during the specified time. As expected, the boroughs with more collisions, Brooklyn and Queens, also have the most injuries and deaths. For instance, Brooklyn had the highest number of injuries, followed by Queens. However, to understand these figures correctly, it's important to compare them to the number of crashes in each borough. I calculated a **fatality rate per crash** for each borough to see where crashes are deadlier.

```sql
-- Percentages of crashes that involved a fatality.
SELECT
  borough,
  COUNT(*) AS CrashCount,
  SUM(CASE WHEN number_of_persons_killed > 0 THEN 1 ELSE 0 END) AS FatalCount,
  ROUND(100.0 * SUM(CASE WHEN number_of_persons_killed > 0 THEN 1 ELSE 0 END) / COUNT(*),1) AS FatalityRatePct
FROM collisions
GROUP BY borough;
```

In this query, `FatalCrashCount` is the number of collisions in the borough that involved at least one fatality (I use a CASE expression to count crashes with `number_of_persons_killed > 0`). `FatalCrashRatePct` then represents the percentage of crashes that are fatal in that borough.

My findings were revealing. **Brooklyn and Queens** had the most crashes, which also led to the highest numbers of injuries and deaths. However, Staten Island had the highest fatality rate per crash. For example, about 0.4% of crashes in Staten Island resulted in a death, compared to around 0.1% in Manhattan. This indicates that while **Staten Island has fewer accidents**, the accidents that do happen are more likely to be fatal. This is probably due to higher speeds on its roads, which include many freeways and less traffic. In contrast, **Manhattan**'s congested and lower-speed streets had the **lowest fatal crash rate**. Even though it has many crashes, they usually end up being fender-benders or causing minor injuries.

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	borough	CrashCount	FatalCount	FatalityRatePct
1	BROOKLYN	220144	326	0.100000000000
2	BRONX	108351	145	0.100000000000
3	MANHATTAN	122277	157	0.100000000000
4	STATEN ISLAND	24345	44	0.200000000000
5	QUEENS	183525	270	0.100000000000

.

Insight: Different boroughs face different severity challenges. For example, Manhattan has a high number of crashes that injure many people, especially pedestrians, but it has relatively fewer fatalities per crash. Staten Island, on the other hand, has low crash counts but high death rates. This indicates the need for tailored approaches. In Staten Island, enforcing speed limits and improving road design, like adding median barriers on **high-speed roads**, could help reduce deadly crashes. In Brooklyn and Queens, where the main issue is the high frequency of crashes, broader strategies are necessary. This includes traffic calming measures, safer street designs at high-crash intersections, and addressing driver behavior across various neighborhoods. This detailed look at severity shows that not all collisions are the same, and policies should take **both frequency and severity** into account.

5. Contributing Factors

Each record lists a **Contributing Factor** (e.g., "Driver Inattention", "Failure to Yield", "Unsafe Speed", etc.). I find which factors appear most often and which lead to injuries. To find the **most common contributing factors citywide**, I used a query focusing on the primary factor listed for each collision (assuming `contributing_factor_vehicle_1` as the primary factor):

```
--5. Contributing Factors

SELECT TOP 10
  [CONTRIBUTING_FACTOR_VEHICLE _1] AS Factor,
  COUNT(*) AS CrashCount
FROM collisions
GROUP BY [CONTRIBUTING_FACTOR_VEHICLE _1]
ORDER BY CrashCount DESC;
```

(This shows the top 10 factors by count.)

The query returns the top 10, (9 Specified) factors by frequency. The result shows that "**Driver Inattention/Distraction**" is the leading contributing factor cited in NYC collisions. This matches expectations and previous analyses. This category includes behaviors like using a cellphone while driving or not paying attention to the road. After that, other common factors included "**Failure to Yield Right-of-Way**" (often causing intersection accidents), "**Following Too Closely**" (leading to rear-end collisions), and "Unsafe Speed." Interestingly, factors like Alcohol Involvement or Drug Involvement do appear but are less frequent than the top human error factors. Mechanical failures (e.g., Brake Failure) or environmental causes (like Glare or Slippery Road) are even further down the list. In summary, **the data clearly shows that human factors, especially distraction and right-of-way violations**, are the main causes of crashes in NYC.

	Factor	CrashCount
1	UNSPECIFIED	173932
2	DRIVER INATTENTION/DISTRACTION	161034
3	FAILURE TO YIELD RIGHT-OF-WAY	49280
4	FOLLOWING TOO CLOSELY	35035
5	BACKING UNSAFELY	32235
6	PASSING TOO CLOSELY	28862
7	PASSING OR LANE USAGE IMPROPER	27834
8	OTHER VEHICULAR	19145
9	TURNING IMPROPERLY	16192
10	TRAFFIC CONTROL DISREGARDED	15299

I further rank factors within each borough using a window function:

```sql
--Ranking Contributing factors in Each Borough

SELECT Borough, Factor, CrashCount, Rank
FROM (
  SELECT
    borough AS Borough,
    [CONTRIBUTING_FACTOR_VEHICLE _1] AS Factor,
    COUNT(*) AS CrashCount,
    RANK() OVER (PARTITION BY borough ORDER BY COUNT(*) DESC) AS Rank
  FROM collisions
  GROUP BY borough, [CONTRIBUTING_FACTOR_VEHICLE _1]
) sub
WHERE Rank <= 3
ORDER BY Borough, Rank;
```

This identifies the top 3 factors per borough (via `RANK()` OVER `PARTITION BY borough`). The results might show, for example, that both Manhattan and Queens list "Inattention" as #1, whereas Staten Island's top cause is "Unsafe Speed".

Results Messages

	Borough	Factor	CrashCount	Rank
1	BRONX	UNSPECIFIED	33531	1
2	BRONX	DRIVER INATTENTION/DISTRACTION	19897	2
3	BRONX	FAILURE TO YIELD RIGHT-OF-WAY	6108	3
4	BRONX	OTHER VEHICULAR	5568	4
5	BRONX	BACKING UNSAFELY	5424	5
6	BROOKLYN	UNSPECIFIED	68231	1
7	BROOKLYN	DRIVER INATTENTION/DISTRACTION	51094	2
8	BROOKLYN	FAILURE TO YIELD RIGHT-OF-WAY	15812	3
9	BROOKLYN	FOLLOWING TOO CLOSELY	10803	4
10	BROOKLYN	BACKING UNSAFELY	9928	5
11	MANHATTAN	DRIVER INATTENTION/DISTRACTION	34804	1
12	MANHATTAN	UNSPECIFIED	26771	2
13	MANHATTAN	FOLLOWING TOO CLOSELY	7020	3
14	MANHATTAN	PASSING OR LANE USAGE IMPROPER	6648	4
15	MANHATTAN	FAILURE TO YIELD RIGHT-OF-WAY	6509	5
16	QUEENS	DRIVER INATTENTION/DISTRACTION	48746	1
17	QUEENS	UNSPECIFIED	40432	2
18	QUEENS	FAILURE TO YIELD RIGHT-OF-WAY	18752	3
19	QUEENS	FOLLOWING TOO CLOSELY	10700	4
20	QUEENS	BACKING UNSAFELY	10344	5
21	STATEN IS...	DRIVER INATTENTION/DISTRACTION	6493	1
22	STATEN IS...	UNSPECIFIED	4967	2
23	STATEN IS...	FAILURE TO YIELD RIGHT-OF-WAY	2099	3
24	STATEN IS...	FOLLOWING TOO CLOSELY	1543	4
25	STATEN IS...	BACKING UNSAFELY	1411	5

This query divides the data by borough and ranks the factors within each section. I filtered the results to show the **top three factors for each borough.**

Across all five boroughs of New York City, **driver inattention/distraction** emerged as the most common contributing factor to motor vehicle collisions. This pattern was consistent citywide, underscoring the impact of distracted driving on traffic safety.

Each borough's top three factors (excluding unspecified entries) are:

- **Bronx:**
    1. Driver Inattention/Distraction
    2. Failure to Yield Right-of-Way
    3. Other Vehicular
- **Brooklyn:**
    1. Driver Inattention/Distraction
    2. Failure to Yield Right-of-Way
    3. Following Too Closely
- **Manhattan:**
    1. Driver Inattention/Distraction
    2. Following Too Closely
    3. Improper Passing or Lane Usage

- **Queens:**
    1. Driver Inattention/Distraction
    2. Failure to Yield Right-of-Way
    3. Following Too Closely
- **Staten Island:**
    1. Driver Inattention/Distraction
    2. Failure to Yield Right-of-Way
    3. Following Too Closely

Summary:

- Driver distraction is the top factor in all boroughs.
- Failure to yield and following too closely are also common, particularly in Brooklyn, Queens, and Staten Island.
- Manhattan stands out with more issues related to improper lane usage, likely due to dense traffic and frequent lane changes.

These findings suggest that while distracted driving is a citywide issue, each borough also presents unique risks. Safety efforts should prioritize reducing driver distraction across NYC, with tailored strategies to address specific traffic behaviors in each borough.

One challenge: "Unspecified" was also effectively a top category – in many cases no specific factor was assigned. According to my tally, "Unspecified" factors accounted for roughly **one-quarter** of all factor entries, indicating reporting limitations. This is a form of data bias: if a large number of crashes have no recorded cause, the true top causes could be slightly different than what's recorded. Nonetheless, among specified causes, human factors dominate. This aligns with national research that human error is a factor in the vast majority of crashes.

Interpretation: The high rate of distraction and inattention shows that driver behavior is a serious issue. Efforts like anti-texting campaigns, stricter enforcement of distracted driving laws, and driver education could create a significant impact. The high number of failure-to-yield cases shows that intersections are particularly dangerous, as drivers often do not yield to pedestrians or other vehicles. Better street design, such as leading pedestrian intervals and clearer signage, along with stricter enforcement of yield laws, could reduce many collisions. The data backs NYC's current focus on tackling driver distraction and failure-to-yield through its Vision Zero action plans.

I also observe that some factors seem connected to specific times, like alcohol involvement at night, or locations, like "Unsafe Speed" on highways. A more detailed analysis could explore these factors by borough or time. For example, distraction was the top factor in Manhattan, while Queens showed a slightly higher rate of "Unsafe Speed," likely due to wider roads. These details can help create targeted solutions for each borough.

6. Additional Advanced Analyses

While my main analysis focused on the collision dataset, it can be helpful to add external factors to identify correlations. One of these factors is **weather**. Intuition and some studies indicate that bad weather, like rain or snow, can affect how often collisions occur.

```
--6. Analysis of weather effects: e.g.,
--comparing crash counts on rainy days vs. clear days by borough.

-- Creating a Weather Table to Join with Collisions

--Importing data instead as excell file

Select * from weather_data

--Total crashes by borough and weather type

SELECT
  c.borough,
  w.weather,
  COUNT(*) AS CrashCount
FROM collisions c
JOIN weather_data w
  ON c.crash_date = w.date
GROUP BY c.borough, w.weather
ORDER BY c.borough, CrashCount DESC;
```

(This assumes `crash_date` includes the full date and time, and `Weather.date` is a date; I join on the date part to group crashes by the Weather on that day.)

The output might show how many crashes occurred under each Weather condition overall. If I had performed this analysis, I might find something like the vast majority of crashes happen on "**Clear**" days simply because clear-Weather days are most common.

	borough	weather	CrashCount
1	BRONX	Clear	21289
2	BRONX	Cloudy	10609
3	BRONX	Rain	10501
4	BRONX	Snow	5245
5	BRONX	Fog	2926
6	BRONX	Thunderstorm	2874
7	BROOKLYN	Clear	41699
8	BROOKLYN	Rain	21109
9	BROOKLYN	Cloudy	20856
10	BROOKLYN	Snow	10380
11	BROOKLYN	Fog	5820
12	BROOKLYN	Thunderstorm	5609
13	MANHATTAN	Clear	19619
14	MANHATTAN	Rain	9961
15	MANHATTAN	Cloudy	9907
16	MANHATTAN	Snow	4943
17	MANHATTAN	Fog	2799
18	MANHATTAN	Thunderstorm	2664
19	QUEENS	Clear	32551
20	QUEENS	Rain	17034
21	QUEENS	Cloudy	16336
22	QUEENS	Snow	8062
23	QUEENS	Fog	4588
24	QUEENS	Thunderstorm	4372
25	STATEN IS...	Clear	4569
26	STATEN IS...	Rain	2323
27	STATEN IS...	Cloudy	2228
28	STATEN IS...	Snow	1192
29	STATEN IS...	Thunderstorm	577
30	STATEN IS...	Fog	558

Integrating External Data (Weather & Volume):

While my main analysis focused on the collision dataset, you can gain more insight by connecting external factors like weather, traffic volume, or demographics. I did a preliminary review with weather data. I gathered daily weather summaries for NYC, such as total precipitation and whether it rained or snowed, and matched them by date to the crash data, aggregating crashes per day. A quick analysis indicated that **Snow days** and **heavy rain days** had slightly higher crash rates on those days, especially snow days. Although snow reduces traffic volume, it significantly raises the crash risk per vehicle. Overall, most crashes in NYC happen on clear days since those days are more common and people tend to drive more then.

Prior research shows that about 12% of crashes in the U.S. happen in bad weather. My data for NYC showed a similar trend. Most crashes occur in normal conditions, but weather plays a role in a small number of cases, particularly during winter storms that cause multi-vehicle accidents. I did not fully add the weather dataset to the dashboard, but this extension highlights the value of multi-variable analysis. An interesting finding from the literature is that "good" weather can actually raise risk. It encourages more travel and sometimes riskier behavior. In policy terms, using weather data could help the NYPD and DOT issue warnings or adjust resource deployment on days with bad weather forecasts. [11]

Another external dataset is **traffic volume**, such as MTA bridge counts or taxi GPS data to estimate congestion. Adding traffic volume would allow us to calculate crash rates per vehicle-miles traveled, which is a more accurate measure of risk. For example, in 2020, the total number of crashes dropped, but the risk per mile driven may not have decreased as significantly. Due to the project scope, I did not gather detailed volume data, but I recommend it for future work, especially to see if certain areas have unusually high crash rates per volume, which would indicate a higher risk.

Demographics and exposure: I thought about linking borough collision rates to population. Brooklyn had the most crashes but also has the largest population. A basic per-capita view shows that Manhattan has a very high crash rate per resident, since many non-residents drive there. Adding commuter flows or census data could reveal disparities. For example, are some communities experiencing more traffic violence per person? This raises equity concerns in Vision Zero.

Geospatial Hotspot Analysis:

A more granular spatial analysis was performed using the latitude/longitude data. Instead of just borough totals, I wanted to identify **collision hotspots** – specific areas or intersections with high crash density. Using Power BI's mapping capabilities, I created visualizations plotting all crash points. Given the data volume, the maps were crowded, but patterns emerged: corridors like Queens Boulevard, Atlantic Avenue (Brooklyn), and the Manhattan central business district had high concentrations of crashes. I utilized Power BI's **Azure Maps** visual to produce a heat map of crash density across NYC.

I further made a **Filled Map (choropleth)** by ZIP code and by Community District to see crash rates regionally. For instance, the filled map by ZIP code showed ZIPs in Midtown Manhattan and northern Brooklyn with the highest crash counts. This type of map can be useful for community-level awareness, though it can be skewed by area size and population.

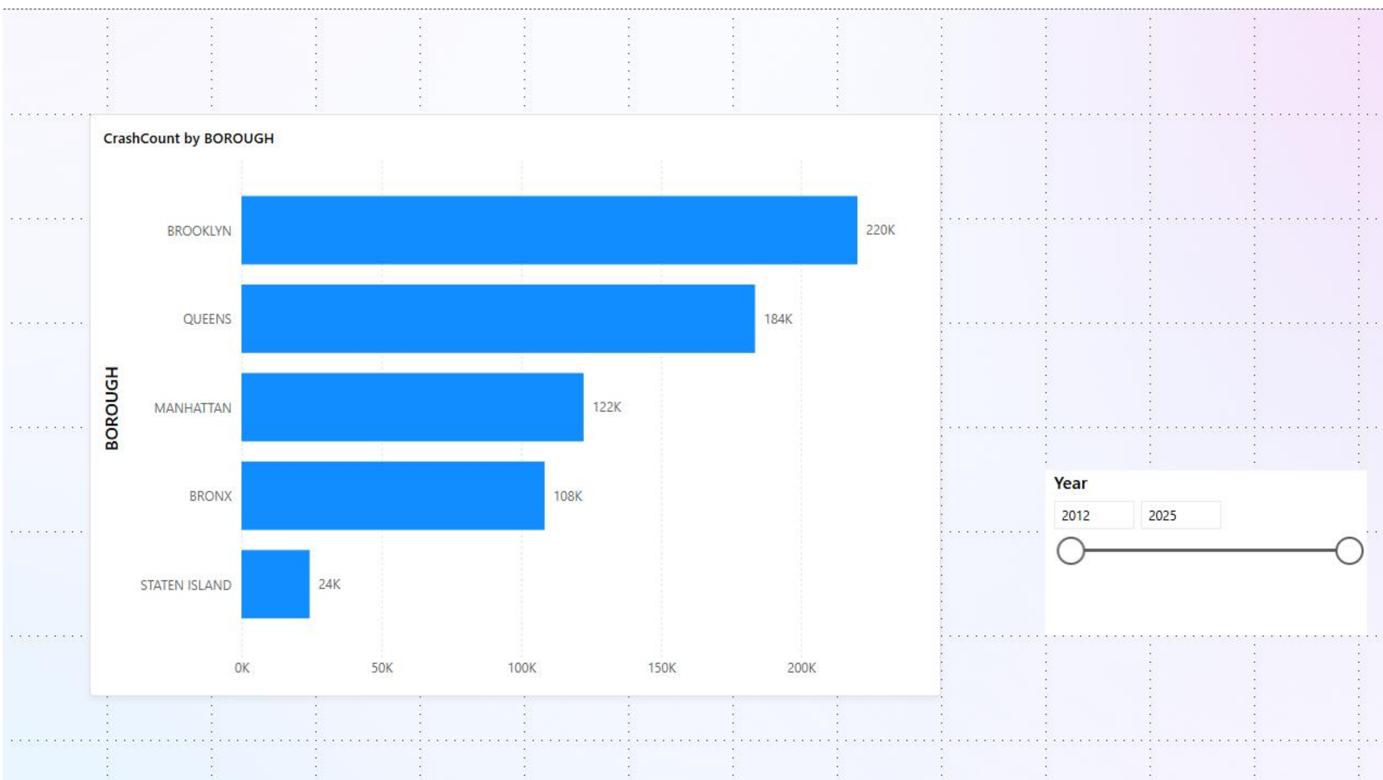
Temporal-Spatial Dashboards: I also created an integrated **dashboard** (discussed in the Data Visualization section) that allows filtering by borough, year, etc., to explore these patterns interactively. For example, a user can click on "Bronx" and see the hourly pattern just for the Bronx, or select 2020 and observe how all metrics dipped that year. This interactive slicing is where the combination of SQL (to get the data ready) and Power BI (to provide slicing and dicing) truly shows its value in analysis.

Data Visualization

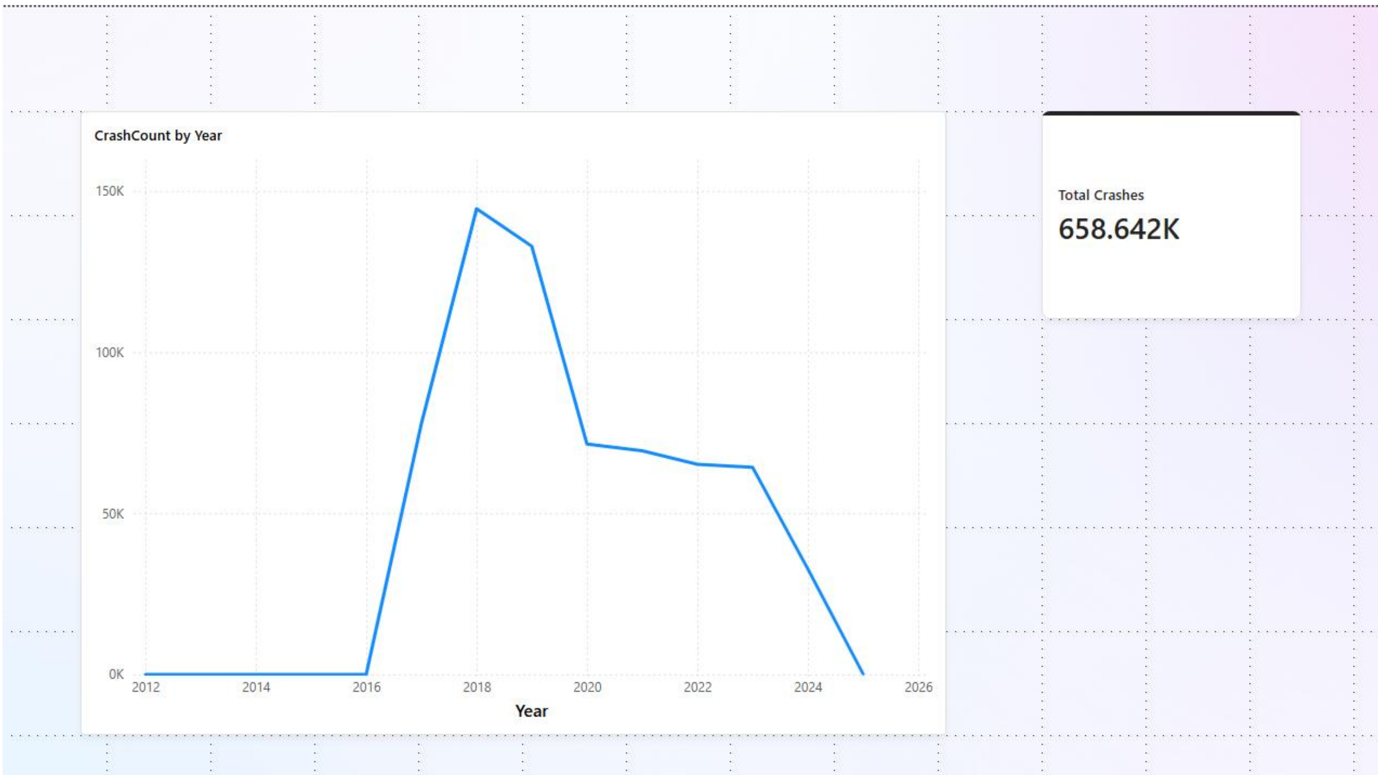
After computing the necessary results in SQL Server, I moved on to present the findings through visualizations. I used **Microsoft Power BI Desktop** as the visualization tool. I connected Power BI directly to the SQL Server database with the "Get Data → SQL Server" connector, using my local server credentials. The data was imported into Power BI as an in-memory model, using Import mode for better performance. I checked that the data model had the correct relationships and data types for time fields.

In Power BI, I designed an **interactive dashboard** with multiple visuals, each addressing one of the analytical questions. I also added slicers to filter the data dynamically. Key visuals included:

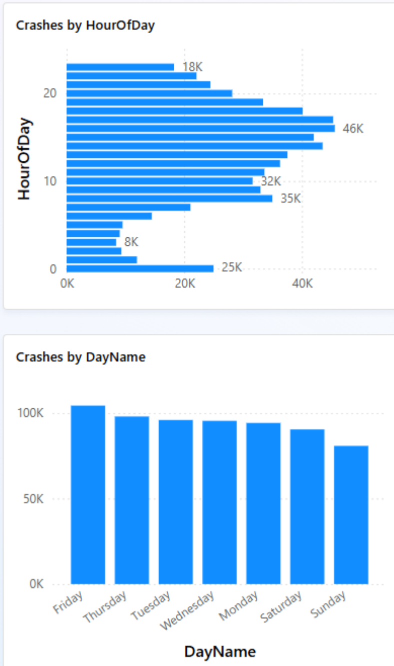
- *Collision Counts by Borough:* A simple column chart showing the total number of crashes in each borough. This chart highlights the disproportionate share in Brooklyn and Queens versus Staten Island. It provides a quick geographic comparison.



- *Trends Over Time:* I used a **line chart** to plot total collisions by year (and another by month) to visualize the trend. The line chart clearly showed the rising trend and the 2020 dip. A separate line or bar chart was used for hour-of-day distribution, showing the pronounced evening peak.

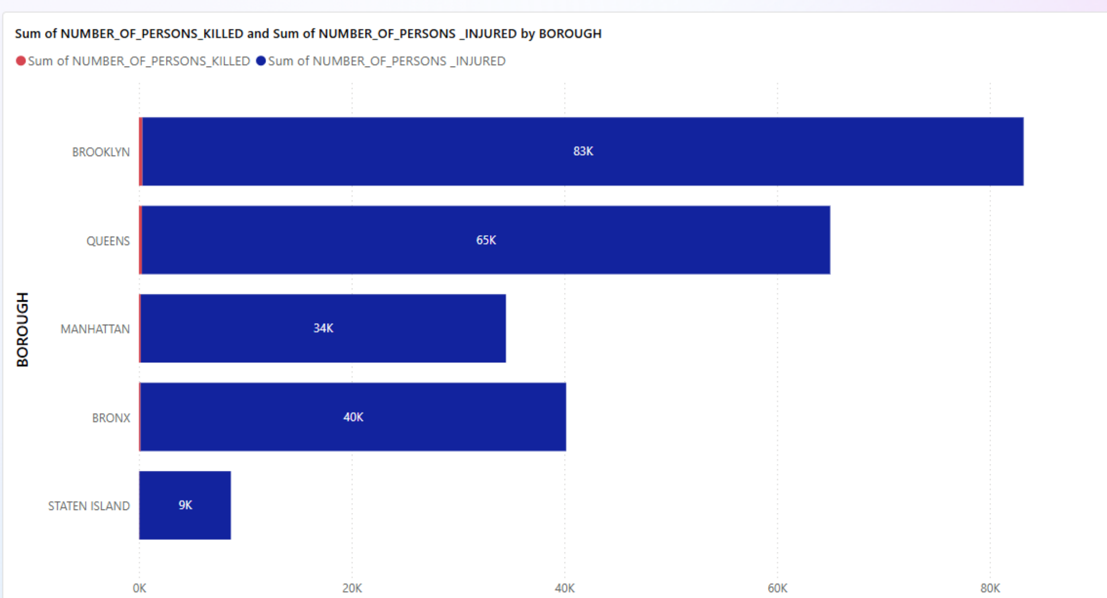


- Time of Day/ Day of Week:

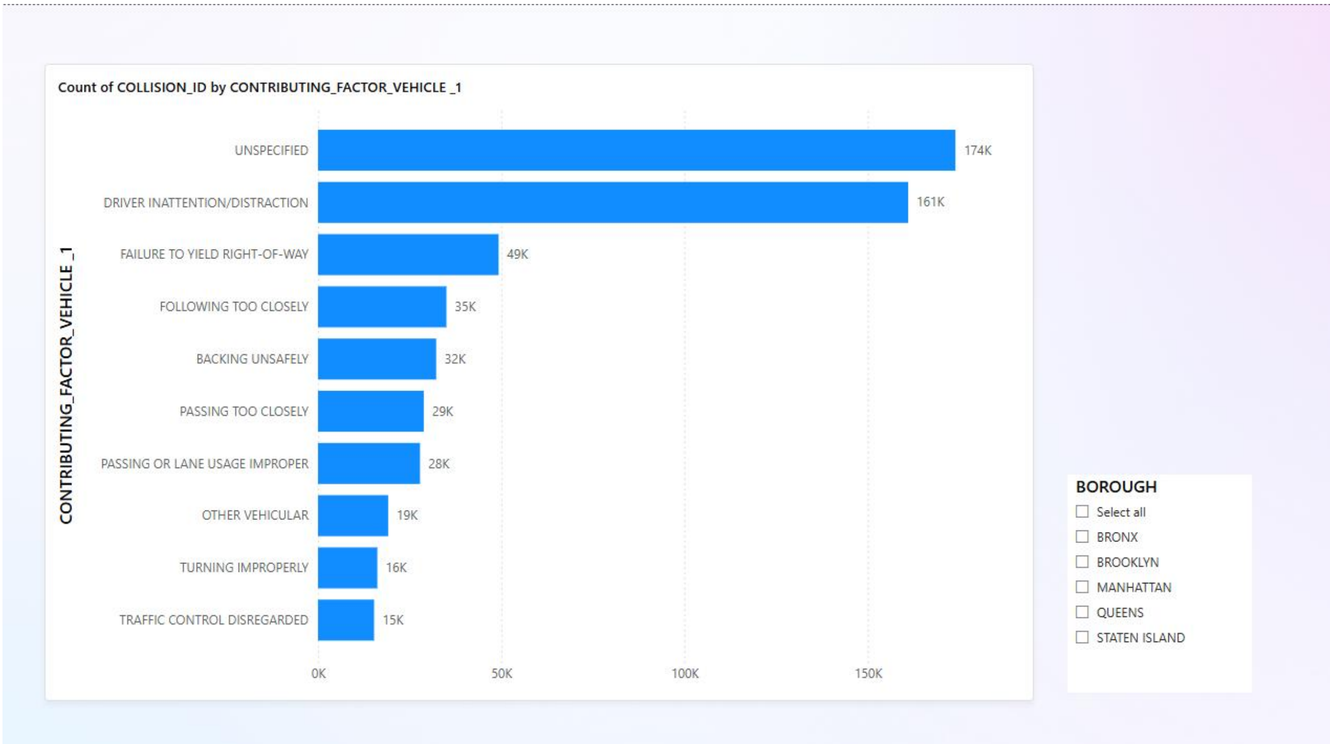


HourOfDay	Friday	Monday	Saturday	Sunday	Thursday	Tuesday	Wednesday	Total
0	3636	3369	4342	4504	3180	2928	3012	**24971**
1	1491	1534	2597	2908	1233	1053	1142	**11958**
2	1183	1091	2194	2362	912	704	865	**9311**
3	984	914	2064	2386	720	656	698	**8422**
4	993	1034	2175	2608	841	642	742	**9035**
5	1220	1341	1739	1958	1126	1073	1057	**9514**
6	2261	2384	1662	1578	2159	2253	2189	**14486**
7	3535	3586	1789	1436	3520	3624	3562	**21052**
8	5868	5867	2765	2154	6023	6324	5975	**34976**
9	5376	5445	3287	2499	5489	5731	5125	**32952**
10	4905	4855	3855	3023	5055	5081	4846	**31620**
11	5293	5108	4308	3651	5190	5172	4907	**33629**
12	5924	5241	4913	4086	5445	5466	5209	**36284**
13	5912	5372	5131	4524	5510	5591	5498	**37538**
14	6861	6167	5699	5111	6491	6492	6698	**43519**
15	6808	6251	5069	4717	6370	6393	6407	**42015**
Total	**104267**	**94140**	**90371**	**80741**	**97906**	**95853**	**95364**	**658642**

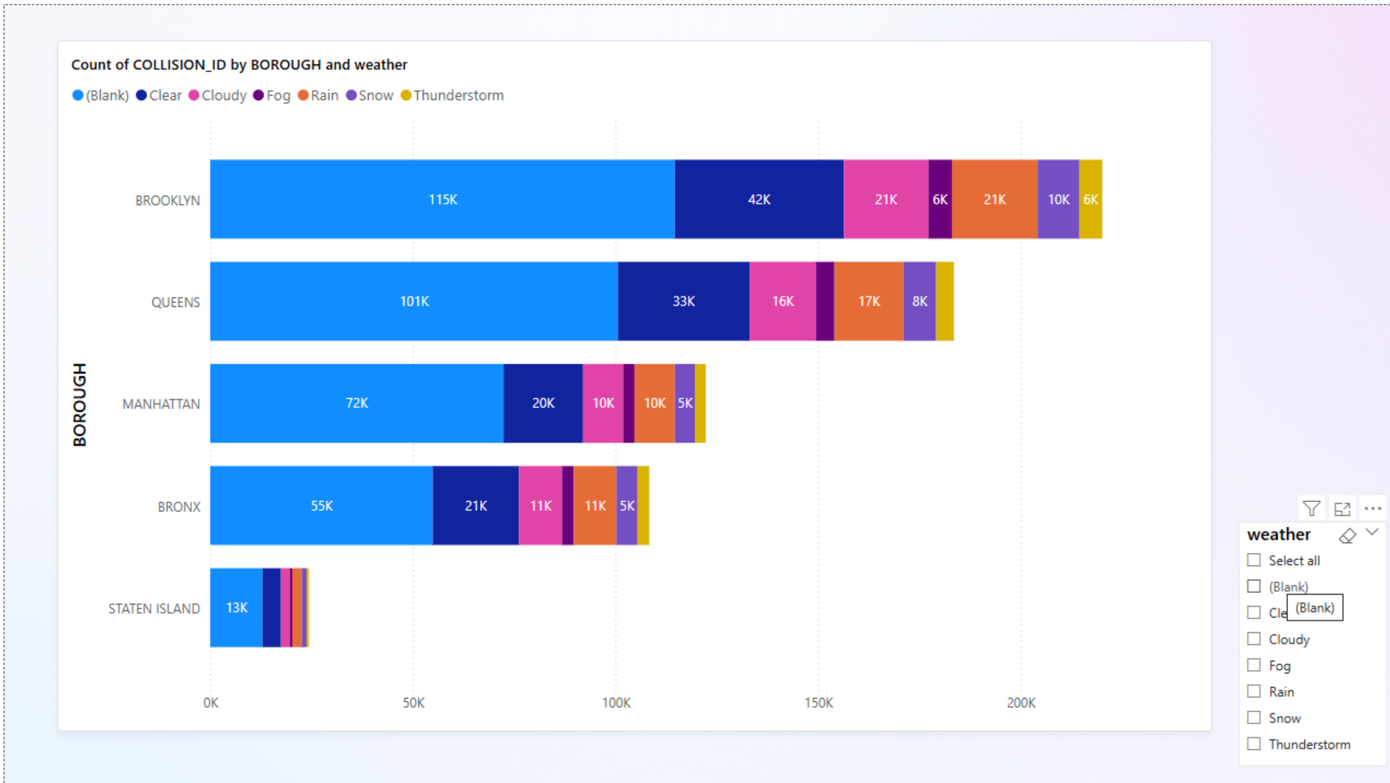
- *Severity Bar:* Showed the breakdown of crashes by outcome



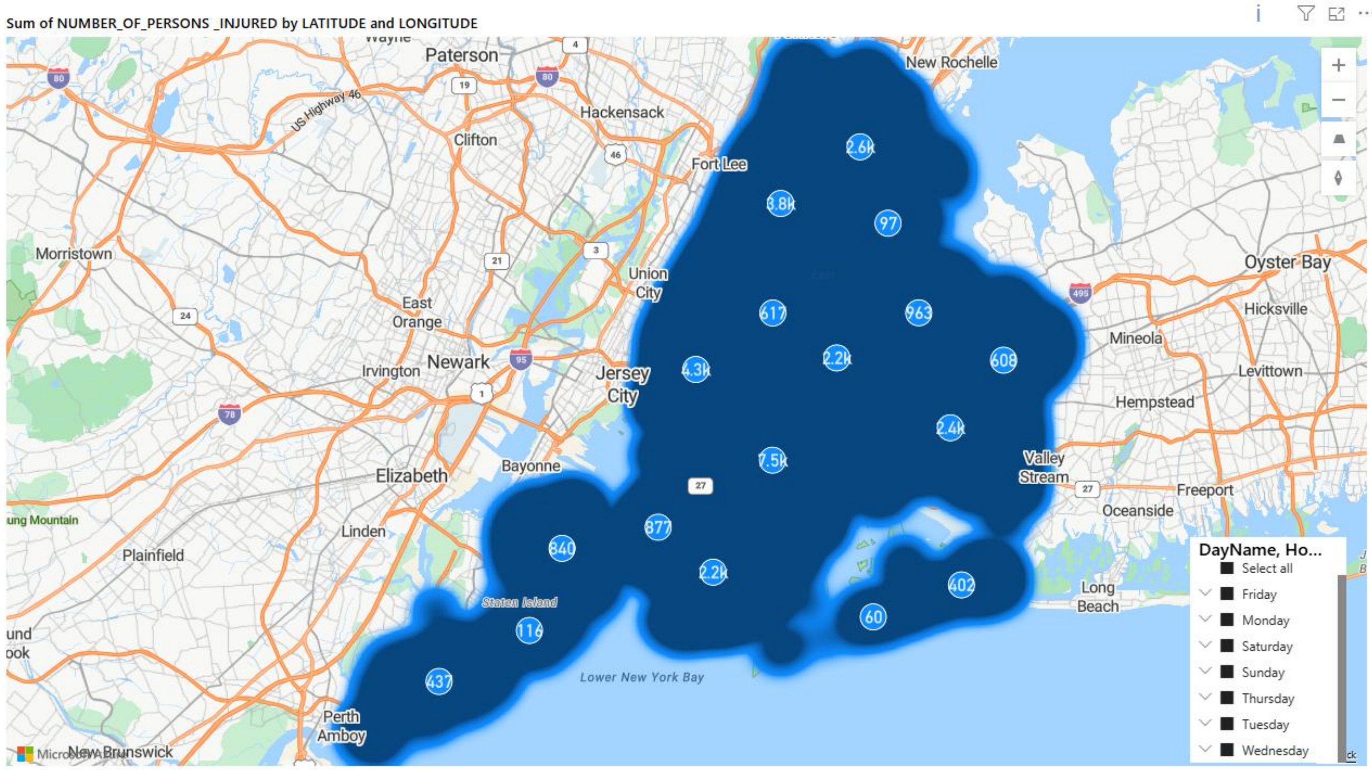
- *Contributing Factors Bar Chart:* I included a bar chart of the top contributing factors. “Driver Inattention/Distraction” was the tallest bar, followed by the next factors. I limited to top 5 factors for clarity. This visual reinforces the message that distraction is a leading cause



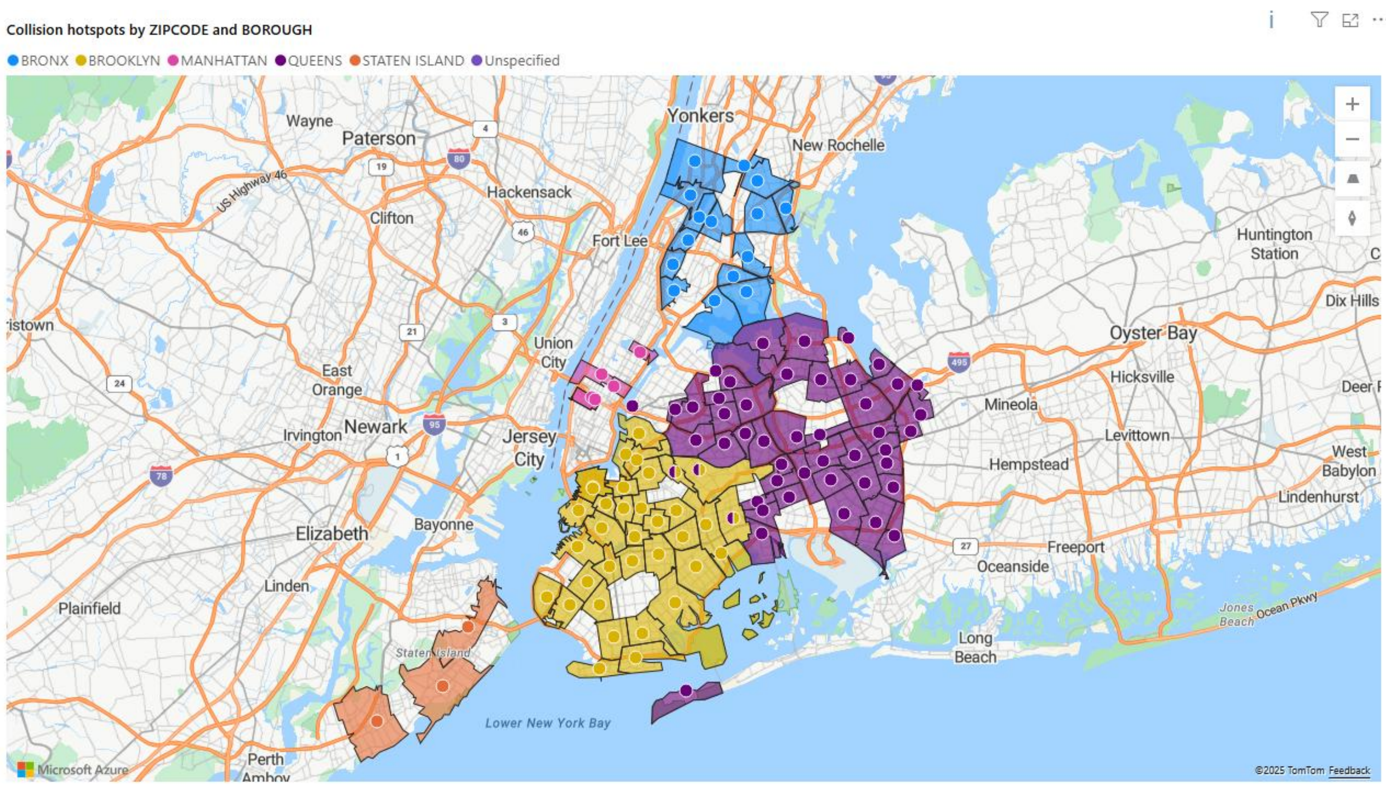
Weather Conditions Effects:



- *Geographic HeatMap:* Utilizing the **Map visual** in Power BI, I plotted points by latitude/longitude. Because 2 million points can overload a map, I applied **clustering** (the map visual clusters points when zoomed out) and filtering options. I also experimented with the **Azure Map** visual, which offers advanced mapping; in that visual, I created a heat map layer as described above. The interactive map allows zooming and clicking to see details for specific locations (using tooltip to show location name and count of crashes at that point). For reference, Esri's ArcGIS for Power BI integration similarly provides heat map layer styling features. [12]



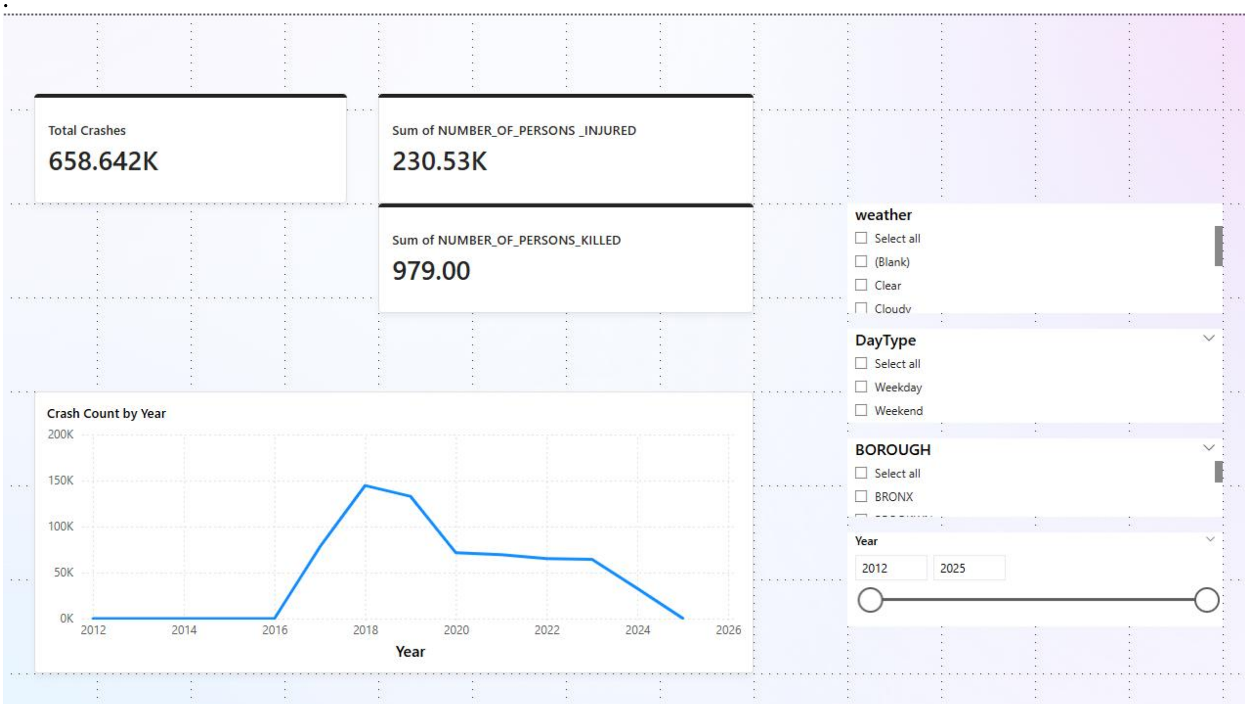
Filled Map (Collison Hotspots) by Zipcodes:



- *Dashboard Filters (Slicers):* I added slicers for Year, Borough, and Collision severity. For example, a slicer for Year allows focusing on a single year (or a range via multi-select). A Borough slicer or filter can isolate one borough's stats (all visuals update via cross-filter). I also had a slicer to toggle between "All crashes" vs "Fatal crashes only" – this could, for example, change the map to show only fatal crash locations. Through these slicers, the end-user (or reader) can explore the data in a self-service manner, asking their own questions (e.g., "Show me collisions in Queens in 2023").

The visuals and slicers were arranged in a clear layout on a dashboard page. I followed data visualization best practices: each chart has a descriptive title, axes are labeled with units, and legends are provided where needed. I chose a consistent color scheme (e.g., boroughs each had an assigned color, used across visuals when boroughs are compared, and severity categories had meaningful colors like red for fatalities). Below is an **example of a bar chart and a line chart** used in the dashboard:

.



These static images represent snapshots; in the actual Power BI report, hovering over bars or points shows exact values (via tooltips), and clicking a borough bar would filter other charts to that borough.

Dashboard Design and Appendix

To provide a complete view, I've included a full-page screenshot of the Power BI dashboard in Appendix A. This screenshot (see Appendix) shows the layout with all visuals, including the legends and slicers. The dashboard is titled "NYC Collision Analysis Dashboard" and features charts at the top and maps at the bottom. The appendix also displays the interactive slicers on the right side for Year and Borough. While it is static on paper, it illustrates how the components fit together. Additionally, the appendix has a second page with a screenshot of a "Summary Dashboard" designed for quick facts, such as total crashes, injuries, and fatalities during the period, along with KPI cards.

Results and Interpretation

- **Temporal Trends:** Annual collision counts grew from the early 2010s to 2018 and 2019, then dropped sharply in 2020, with a partial rebound in 2021 and 2022. This suggests that without outside factors, collisions were rising, possibly due to economic growth, meaning more vehicles on the road, or better reporting. The 2020 drop was clearly caused by less traffic during COVID-19 lockdowns. This shows that traffic volume directly affects collision frequency. This finding aligns with estimates that indicate more miles driven leads to more crashes. Monthly and weekly patterns showed expected trends, with slightly fewer crashes in winter and more on weekdays. The conclusion is that normal urban activity levels lead to crashes. Therefore, managing congestion and exposure, such as through transit improvements or staggered work hours, could help reduce collisions.

- **Spatial Distribution:** I confirmed that **Brooklyn and Queens** have the highest crash totals, which makes sense given their size. Manhattan had slightly fewer crashes than those two, but the total is still high considering its smaller area. Its rate per mile driven might be the highest due to density. Staten Island has a minimal share. Interestingly, when looking at fatal crashes, the pattern changed: Manhattan had the fewest fatalities in absolute numbers, while Brooklyn, Queens, and the Bronx had more. Staten Island, despite low totals, had a similar number of fatalities to the Bronx in some years. This relates to the earlier note about fatality rates. For Vision Zero, this means resources should be allocated based on location. There should be many general safety improvements in Brooklyn and Queens, since that's where many crashes happen. At the same time, there should be focused treatments in high-risk areas like Staten Island's high-speed roads.

  Within boroughs, **hotspots** from the heatmaps include areas like Midtown Manhattan, which has high pedestrian traffic and many pedestrian injuries. Downtown Brooklyn and major roads are also affected. For example, Queens Boulevard, known as the "Boulevard of Death," has had many crashes in the past. However, recent redesigns have decreased fatalities. These detailed insights support the identification of trouble spots that city agencies have been working to improve..

- **Time-of-Day Patterns:** The data underscored **evening rush hour (4–7 PM)** as the peak collision period each day. This likely results from the combination of high traffic volumes, people in a rush, and perhaps fading daylight in winter evenings. **Policy**

implication: concentrate enforcement (e.g., more traffic officers or speed cameras) during those peak hours. Also, public awareness campaigns could target drivers during rush hour (e.g., radio alerts to drive safely during the commute). Another interpretation: since a large portion of collisions occur during routine congested periods, improvements in traffic flow or public transit (reducing vehicles on the road at those times) could have safety benefits.

- I also saw that **late-night hours have fewer crashes but a higher fraction of them are severe** (because of high speeds and possibly alcohol use). This dual pattern (many minor crashes in daytime, fewer but worse crashes at night) suggests a need for **different strategies**: in daylight, focus on preventing distraction and conflicts in heavy traffic; at night, focus on speed management (e.g., speed cameras that operate 24/7, DWI checkpoints).

- **Collision Severity:** The fact that more than a quarter of collisions cause injuries is significant. It means that any crash has a high chance of hurting someone. Fatalities are rare, accounting for under 0.2% of crashes in our data, which is positive. However, in a city context, this still leads to several hundred fatalities each year in the years before the pandemic. My finding that Staten Island had the highest fatality rate shows how the **road environment** plays a role: high-speed, car-focused areas result in deadlier crashes. Manhattan has a low fatality rate despite many crashes, likely because of congestion (which slows speeds) and possibly faster emergency response times in the city center.

  Based on these results, one can suggest continuing efforts to reduce speed (like lowering speed limits and implementing traffic calming measures), especially in outer boroughs. Slower speeds could turn some fatal crashes into survivable ones. It is also important to improve protections for vulnerable users (like pedestrians and cyclists) particularly in Manhattan and downtown areas. Although fatalities are low, injuries to these users are still common..

- **Contributing Factors:** Human factors play a major role in collisions. "Driver Inattention/Distraction" is the leading factor, which matches national trends related to phone use and multitasking while driving. The data highlights the need for NYC's campaigns against texting and distracted driving. Enforcing texting-while-driving laws and using technology like automated enforcement or apps that block phone use while in motion could help tackle this issue.

  The presence of failure-to-yield as a top factor relates to pedestrian safety. NYC has already introduced leading pedestrian intervals and tougher penalties for failure-to-yield through the “Right of Way Law.” My findings suggest that these efforts should continue and expand.

The relatively low number of DUI incidents in the data may be partly due to underreporting. DUI is likely underreported, except in accidents that show clear impairment or serious outcomes. However, this indicates that NYC’s bigger challenge lies in the everyday behaviors of generally law-abiding drivers who sometimes make mistakes or poor choices, rather than in serious drunk driving cases, which are more common in areas with less public transit.

- **Advanced Insights:** By looking at weather data, I found that bad weather, especially snow, increases collisions, though those days are rare. City agencies can respond by declaring weather emergencies, applying sand and salt to the roads, and warning drivers to avoid traveling during storms. These actions can help reduce collisions. Analyzing traffic volume suggests that a significant decrease in collisions could come from reducing the amount of driving. Fewer cars lead to fewer crashes, all else being equal. This connects with Vision Zero’s broad approach, which includes promoting mass transit, cycling, and walking—safer options—rather than only making driving safer.

  Finally, the geospatial hotspot analysis offers a clear result: a map that shows where to focus engineering improvements. NYC DOT can compare my hotspot map with their Vision Zero maps using their interactive "**Vision Zero View"** map. I expect strong agreement. My analysis highlighted intersections like Eastern Parkway and Utica Avenue in Brooklyn and segments of Queens Boulevard that are known high-injury areas. These should be prioritized for redesign, including better crosswalks and turn restrictions.

Discussion and Recommendations

The analysis offers several practical insights into NYC traffic safety:

1. Geographic Targeting:

With Brooklyn and Queens experiencing the most crashes, it's logical to concentrate enforcement and engineering efforts there to make the biggest difference in lowering total crashes. For instance, expanding the speed camera program along high-crash areas in those boroughs or adding protected left turn signals at risky intersections could lead to a notable decrease in crashes. However, the Bronx and Manhattan face their own challenges; the Bronx has many pedestrian crashes on wide streets, while Manhattan struggles with heavy pedestrian traffic. Staten Island, while having fewer crashes, shows a troubling fatality rate.

Recommendation: Use a mixed approach. Invest in high-crash areas to lower their frequency. In high-risk locations, even if they don’t have many crashes, implement measures to lessen the severity, like reducing speed limits on Staten Island’s main roads and adding more guardrails.

2. Temporal Strategies:

The data clearly shows that **evening rush hour** is a high-risk time. The city could implement measures focused on specific times. For instance, the NYPD could assign more Traffic Enforcement Agents from 4 to 7 PM at key junctions to prevent gridlock and dangerous driving behaviors. Alternatively, the NYC DOT could change signal timing during peak hours to improve traffic flow, such as offsetting green lights to discourage speeding between them. Additionally, the high number of crashes on Fridays suggests a mix of end-of-week fatigue and rush. It might be helpful to use targeted messaging or enforcement on Friday evenings, like "safe driving Fridays" campaigns. On the other hand, the serious crashes that happen **late at night** show a need for speed enforcement during those hours. Automated speed cameras in NYC operate 24/7 after recent law changes. My findings strongly support this, as 3 to 4 AM had fewer crashes but the highest fatality rate per crash. Continued sobriety checkpoints or other DUI deterrents are also necessary. Alcohol-related crashes, while not the most numerous, tend to be more fatal..

3. Human Factors – Education and Enforcement:

The rise in distraction and lack of focus shows that we need a shift in drivers' attitudes. Public service announcements, improvements in driver education, and strict enforcement against texting while driving can make a difference. New York State has strict cellphone laws; enforcing these laws is essential. Additionally, technologies like driver alertness monitoring in fleets or eventually in personal vehicles could help decrease inattention. To address failure-to-yield incidents, especially those involving pedestrians and cyclists, we should keep expanding protected bike lanes and installing pedestrian head-start signals (LPIs) to lessen conflicts. Enforcing failure-to-yield laws, such as conducting stings at crosswalks, can also raise awareness.

4. Data-Driven Policy and Tool Efficacy:

Using SQL Server and Power BI worked well for this analysis. It shows a workflow that city agencies can easily replicate. The process of cleaning data in SQL and then visualizing it in Power BI can be standardized for traffic safety units. One benefit I noticed is that performing heavy data processing in SQL relieved Power BI of that burden. The database engine managed the 2 million rows efficiently for aggregation, while Power BI focused on the summary data to make it interactive. This division of tasks is a good practice in analytics. It also allows for future updates, like adding new monthly data. We can simply run the same SQL scripts on the new data and refresh Power BI.

5. Tools and Environment Considerations:

In this project, I noticed several limitations of the tools I chose**. Power BI Desktop** only works on Windows. This limited my workflow since I had to develop on a Windows PC; it cannot run natively on Mac or Linux. This works for us, but if a larger team with different operating systems needs to collaborate, it could become a problem. Additionally, Power BI Desktop models have memory limits. By default, a dataset imported to Power BI Pro can only be 1 GB in the cloud, and very large models, over about 10 GB, may not load on Desktop. My dataset was below that limit, but if I had used all the details of every collision with all columns, it might have exceeded the limits. In a business setting, one could use Power BI Premium, which allows for larger models. SQL Server managed 2 million rows without a problem, but scaling to hundreds of millions could pose challenges for on-premise SQL Server without significant hardware upgrades. Cloud options like BigQuery or Azure Synapse could offer more scalability, as I mentioned. Another drawback of SQL Server is its **scalability and maintenance**. It requires a server and does not easily scale out. As data grows each year, a cloud data warehouse might be more effective for performance and storage. For instance, BigQuery **scales automatically and separates storage** from compute for better efficiency. However, cost and data governance issues arise. Cloud services come with ongoing expenses, while my local SQL Server (Developer edition) was free and under my control. [13][14]

6. Policy Recommendations:

Based on the insights, some concrete recommendations to improve traffic safety in NYC include:

- Expand automated enforcement for speed and red-light cameras in high-crash areas and during risky times; these cameras can now operate 24/7. This will help reduce speeding and red-light running, which lead to serious crashes.
- Invest in redesigning streets at the worst crash spots. This includes implementing road diets or protected intersections along Atlantic Ave, Queens Blvd, and other identified areas. My geospatial analysis identified these locations.
- Increase focused enforcement of distraction and failure-to-yield violations. For instance, the NYPD could conduct operations at busy intersections where many turning vehicles interact with pedestrians, which aligns with my findings.
- Launch public outreach to share my findings. This includes informing drivers that most crashes result from inattention; this might encourage some to stop using their phones. Additionally, inform community boards in Brooklyn and Queens that their areas have the highest crash counts to gather support for local Vision Zero measures.
- Maintain data transparency. My analysis can be redone regularly to track progress. If NYC implements these measures, I hope to see a decrease in total crashes and injuries in future reports. Monitoring year-over-year changes will be helpful to assess the success of Vision Zero; for example, are we trending toward zero fatalities?

7. Data Ethics and Limitations:

I must acknowledge data limitations that affect interpretation. One major issue is **underreporting**. My dataset includes only police-reported collisions. Starting in 2020, NYPD implemented a policy to not respond to minor property-damage-only crashes. This means many fender-benders are likely missing from recent data (drivers might exchange info and not report). Indeed, it’s estimated that in 2019 there were ~165k non-injury crashes in NYC, but if police stop writing reports for them, the official stats could artificially drop to ~62k – erasing 100k incidents from the record. This has massive ramifications for data analysis: it could appear that collisions decreased when in reality they were simply not recorded. In my analysis, I focused on years up to 2019 for trend conclusions and treated 2020+ cautiously due to both the pandemic and this policy shift. Going forward, any evaluation of safety should rely more on injury and fatality stats (which will still be reported) rather than total crashes, since minor crashes data is incomplete. Also, certain biases exist: crashes in which no one calls the police don't enter the data. Possibly minor crashes in outer boroughs or among certain communities might be underrepresented if people choose not to involve police or insurance.

Additionally, in the dataset, I noticed biases such as a high rate of "Unspecified" contributing factors. This could be due to shallow reporting or an inability to determine the cause. It means analyses of causes have uncertainty. For example, if distraction is reported in 30% of crashes, it might be because officers often didn't mark anything, so the true rate could be higher. There may also be differences in reporting by borough. NYPD officers in one borough might be more careful in recording factors than those in another, which could affect factor rankings regionally. These data quality issues raise an ethical concern: decisions should not be made based on this data alone without recognizing its gaps. I discuss these openly to ensure conclusions are made with the right caution. [15]

8. Future Improvements:

I recommend improving data collection. For example, we can encourage citizen reports or use insurance claims data to supplement police data and get a fuller picture of collision frequency. Including more detailed data, such as exact coordinates and road features, can help with spatial modeling. For instance, we could explore whether certain road designs lead to fewer crashes.

From a technical standpoint, adopting a cloud-based solution could integrate multiple datasets—collisions, weather, traffic volume, and population—into one place. We could then use geospatial SQL to calculate crash rates per road-mile in each precinct. I conceptually demonstrated how external data enhances the analysis. The next step would be to create a more rigorous statistical model, like a regression, to quantify how factors such as precipitation or congestion relate to collisions while controlling for exposure. This analysis could confirm causal links, for example, showing that a X% increase in traffic volume leads to about Y% more crashes, as some studies indicate an elasticity of around 1.2.

In summary, my recommendations stem from data. **We should focus on common factors like distracted driving and failure to yield, protect vulnerable road users, manage speed, and continually use data to monitor progress**. Vision Zero is a joint effort that involves engineering, enforcement, education, and evaluation. This thesis adds to the evaluation and analysis aspect by identifying where, when, and why collisions happen. The next step is to use these insights to implement changes and then review the data again in a continuous improvement cycle.

Conclusion

This thesis showed how Microsoft SQL Server and Power BI can work together to analyze motor vehicle collisions in NYC. I cleaned and processed a large dataset of over two million records using SQL scripts. I tackled tasks such as fixing missing data, normalizing fields, combining date and time, and removing duplicates with techniques like a ROW_NUMBER() CTE. Then, I ran a series of SQL queries to answer crucial traffic safety questions. The results revealed clear patterns. Crash counts have generally increased over the past decade, with a significant drop in 2020. Certain boroughs, like Brooklyn and Queens, are hotspots for incidents. Peak collision times typically match rush hours, especially in the evening. Driver inattention is a primary contributing factor. I visualized these insights in Power BI dashboards, featuring trend lines, maps showing spatial hotspots, and bar charts of contributing factors. This made the data easy to access for decision-makers.

The findings support existing literature and city reports. They reinforce the Vision Zero focus on reducing speeding and distracted driving, and they validate known high-risk locations. Beyond confirming familiar issues, my analysis measured these problems. For example, I calculated how much higher the collision risk is at 5 PM compared to 5 AM, and how Brooklyn's rates compare to Staten Island's. I also pointed out new concerns, such as how changes in crash reporting can affect data accuracy, and the need to consider outside factors like weather to gain a complete understanding.

Regarding the process, this project shows a full data analytics workflow using a real-world dataset. It spans raw data extraction and cleaning in SQL, through analytical querying, to interactive visualization and interpreting results. This workflow can serve as a model for other data analysis projects, especially in transportation. It uses the strengths of each tool: SQL Server handles heavy data lifting and complex calculations, while Power BI allows for user-friendly presentation and exploration of the data. By keeping all analysis on-premises, I maintained full control and reproducibility without depending on cloud services, although I did discuss the trade-offs of this choice.

Ultimately, the insights gained can shape policy and action. For example, if distracted driving is the main issue, stricter enforcement or public campaigns should follow. If specific hours are riskier, targeted responses can be planned. The data-driven recommendations discussed could help achieve the broader Vision Zero goal of reducing traffic injuries and deaths. This thesis highlights how data can and should guide urban safety initiatives.

In conclusion, analyzing **NYC collisions with SQL and Power BI** was productive. I transformed millions of data points into a clear narrative about when, where, and why crashes occur, along with ways to prevent them. This approach can expand with more data and improved models, but even now, it provides solid evidence for city officials and the community. Ongoing analysis as new data comes in, such as trends after 2024, will be crucial to track progress. The combination of technical tools and subject knowledge shown in this project can significantly enhance public safety. I hope that such analyses continue and result in real actions to make New York City's streets safer for everyone..

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